

Programming the Internet of Things Using Contiki and Cooja

Contiki is an operating system with a focus on low power IoT devices. Cooja is the Contiki network simulator. Cooja allows the large and small networks of Contiki motes to be simulated. This article takes the reader through the process of programming IoT with Contiki and Cooja.



We are all surrounded by a number of gadgets, mobile devices, smartphones, wireless nodes and many other objects that are digitally connected in real-time. The Internet of Things (IoT) enables real world objects to interact with each other. These objects can share information and communicate in real-time to deliver better performance as well as security. IoT works by developing and integrating smart objects that can be controlled using remote network infrastructure.

The term, the Internet of Things, was coined by Kevin Ashton in 1999. The implementation of IoT is now widespread because of high performance wireless technologies. Radio frequency identification (RFID) tags and sensors are the most important components of IoT. The RFID tags can be embedded in real world devices and objects, which can be monitored remotely using software based applications. RFID readers can be used to locate, read and sense RFID implanted objects. Very small, micro-sized transmitting and receiving chips are integrated with RFID to

help them communicate with distant objects.

As per reports from *Forbes.com*, the market for the Internet of Things is expected to reach around US\$ 267 billion by 2020. Analysis from Gartner underlines that around 8.4 billion objects, with investments amounting to US\$ 273 billion, will be interconnected with each other in the current year (2017-18). This represents a 31 per cent increase over the figures of 2016-17.

Some of the key applications of IoT include:

- Smart cities, retail points, smart grids, agriculture and farming, homes and offices
- The Internet of Vehicles (IoV)
- Connected cars
- Connected railways infrastructure
- Wearable devices
- Software defined networking
- The industrial Internet
- Energy management